

1 Project Overview

- Title: Trade-Offs Between Success-Attack Rate and Bandwidth Overhead for Timing-Based Webpage Fingerprinting Attacks
- Faculty mentor: Rémi A. Chou
- Program track: Classic CSE Dept REU
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- Project summary: While low-latency anonymity systems like Tor [1] encrypt data content, they remain vulnerable to traffic analysis [1, 2]. Specifically, Website Fingerprinting (WF) allows attackers to infer a user's destination by analyzing packet timing, direction, and volume [3, 4]. Defenses like Tamaraw mitigate this by reshaping traffic into uniform patterns through fixed-interval transmissions and size padding [5, 4]. While Tamaraw effectively lowers attack-success percentage (ASP) in multi-webpage open-world settings [5, 3, 6, 7], it is less robust in single-page settings, where ASP can exceed 90 percent[8]. Although subsequent enhancements introduced more randomness and heavier padding to reduce ASP, they incurred substantial Bandwidth Overhead (BO) [8, 9]. This project investigates the trade-off between ASP and BO within the Tamaraw framework. By adapting the algorithms from [8, 9] and utilizing contemporary datasets [6, 10], we will systematically sweep parameters—such as length bucket size and random padding strength—to quantify how much overhead is required to effectively suppress ASP in single-webpage evaluations.

2 Research Problem

Used by two million people daily, Tor is a vital privacy tool [11] increasingly targeted by WF, which is a passive traffic analysis attack [2]. To counter WF, researchers use methods like padding and dummy traffic to lower the ASP, though this inevitably increases the BO.

Currently, there is still lacking a fully systematic framework for understanding the interplay between these two metrics. Our study aims to address this gap by analyzing the optimal trade-offs between ASP and BO for the Tamaraw defense. By focusing on a single-webpage setting, we provide a more stringent and realistic assessment of defense efficacy, reflecting practical threats where an attacker's primary goal is monitoring a specific sensitive webpage.

3 Research Plan

The details of our plan is as follows.

Student work plan:

- Setup and Literature review: Setup the environments for implementation and conduct literature review on the latest progresses in this research area.
- Modify the original Tamaraw code [9] and WF attacks [6] for our analysis purpose. Specifically, configure fixed per-direction rates, length buckets, and optional randomness to generate parameter-tuned Tamaraw settings.

- On a parameter-tuned Tamaraw setting, run one-page attacks for selected monitored pages, analyze results to illustrate trade-off curves (ASP vs BO) and comparisons.
- Package scripts, configs, and notebooks into a reproducible artifact and prepare figures plus concise text for a summary and poster.

Evaluation:

- Analysis: For each page, plot how ASP changes with stronger defense (larger buckets, more randomness) and relate this to BO to find knees and diminishing-returns regions.
- Metrics: ASP and BO for the single monitored page across various parameter-tuned Tamaraw settings.
- Overhead: Bandwidth ratio defined as the size of data with defended scheme over the size of data with undefended scheme (original data).

Methods / Resources:

- Use existing WF processes [6, 8, 9] and dataset [6] where allowed, running on UTA lab/department machines (GPU-focused).

Timeline (12 weeks)

- Weeks 1–4: Set up development environments. Conduct literature review on Tamaraw variants, WF attacks, and recent defenses. Begin initial code familiarization with Tamaraw and attacks.
- Weeks 5–9: Define and run the parameter tuning for a small monitored-page set (more pages if time is enough), collecting ASP and BO and generating plots/tables.
- Weeks 10–12: Refine analysis, highlight key trade-offs and recommended parameters, finalize reproducible artifacts, create a poster and a page summary, and practice presentation.

Feasibility and risks

- Tamaraw details: If mapping from the literature to datasets is impractical, use a simplified Tamaraw-like scheme and clearly document approximations.
- Experiment volume: If full sweeps are too heavy, restrict to a few representative pages and a compact parameter tuning, emphasizing depth over breadth.

4 Intellectual Merit

The obtained results will produce a quantitative map of Tamaraw's trade-offs between ASP (privacy) and BO (cost) in the one-page setting, which highlight parameter ranges that give strong one-page protection without extreme overhead and show how different webpages behave under the same Tamaraw configuration.

5 Outcomes and REU Plan

Final deliverable package: We plan to make a poster summarizing the research background, experimental setups, methodologies, and key trade-offs between ASP and BO results. Also, we plan to link to artifacts (Git repository) containing cleaned code, configuration files, and non-restricted data artifacts or feature matrices along with a brief reproduction guide.

Presentation: We plan to deliver a poster presentation at Innovation Day (April 14, 2026) or another approved venue.

References

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